Archiv für Elektrotechnik 75 (1992) 163–168

Archiv für Elektrotechnik

© Springer-Verlag 1992

Two algorithms of the fundamental harmonic complex RMS value calculation

L. S. Czarnecki, Baton Rouge, USA

Contents: This paper discusses two algorithms for the complex rms value $\mathbf{X}_1$ of the fundamental harmonic of the nonsinusoidal quantities calculation which are much more computationally effective than the traditional FFT algorithms. They mitigate, moreover, the requirement regarding the sampling rate of the A/D converters as compared with the method suggested by Sawicki in [4]. One of these algorithms is based on the DFT direct calculation, the second applies the cyclic convolution. The paper shows also how to evaluate the aliasing error and the error caused by the lack of the sampling frequency synchronization with the frequency of the sampled quantity.

Zwei Algorithmen zur Berechnung des fundamentalen harmonischen Effektivwertes

Übersicht: Diese Abhandlung beschreibt zwei Algorithmen zur Berechnung des complexen Effektivwertes $\mathbf{X}_1$ der fundamentalen Frequenz von nichtsinusförmigen Schwingungen, die einen wesentlich geringeren Rechenaufwand erfordern, als die herkömmlichen FFT Algorithmen. Sie reduzieren außerdem die Forderungen bezüglich der sampling rate des A/D Konverters im Vergleich zur Methode, die von Sawicki [4] vorgeschlagen wurde. Einer dieser Algorithmen basiert auf der DFT Direktberechnung, während der andere die zyklische Faltung verwendet. Außerdem wird beschrieben, wie der aliasing Fehler abgeschätzt werden kann, als auch der Fehler bedingt durch die fehlende Synchronisation der sampling Frequenz mit der Frequenz, der zu diskretisierenden Schwingung.

1 Introduction

The rms value of the voltage and current and their phase-shift in power systems with sinusoidal waveforms provide a substantial information on the system performance. In the case of power systems with a moderate distortion of the voltage and current waveforms a similar information is conveyed by the rms value and the phase-shift of their fundamental harmonic. Their utilization for the system control, protection, and energy accounts is the reason for which these values are the subject of a particular metrologial interest.

The advance in digital methods of data acquisition and its processing enables the information on the rms value of the voltage and current fundamental harmonic and its phase be provided together, in a form of the complex rms (crms) value, defined ($\triangleq$) as

$$\mathbf{U}_1 \triangleq U_1 \, e^{j\alpha_1}, \quad \mathbf{I}_1 \triangleq I_1 \, e^{j\beta_1} \tag{1}$$

or generally

$$\mathbf{X}_1 \triangleq X_1 \, e^{j\xi_1} \triangleq \mathrm{Re}\,\{\mathbf{X}_1\} + j\,\mathrm{Im}\,\{\mathbf{X}_1\} \tag{2}$$

related to the fundamental harmonic $x_1(t)$ of the quantity $x(t)$ with the following convention

$$x_1(t) = \sqrt{2}\,\mathrm{Re}\,\{\mathbf{X}_1 \, e^{j\omega_1 t}\}, \quad \omega_1 \triangleq 2\pi f_1 = 2\pi/T_1. \tag{3}$$

The crms value of the voltage and current fundamental harmonic in systems with strongly distorted waveforms enables us to determine the reactive power Q_1 of the fundamental harmonic, which has a discriminated meaning in power theories suggested by Kimbark [1] and by Depenbrock [2]. Also just this power is minimized with thyristor controlled reactors used for the power factor improvement.

There is also another reason for accurate measurements of the crms values $\mathbf{U}_1$ and $\mathbf{I}_1$. Their value and the active power P of the load value enables us [3] to determine the internal loss of the active power ΔP_s in the originally sinusoidal power system with the waveforms distorted by a non-linear or/and time-variant load. Namely, if $P_1 \triangleq \mathrm{Re}\,\{\mathbf{U}_1\mathbf{I}_1^*\}$ is the load active power of the fundamental hamonic and $\mathbf{Z}_{s1} \triangleq \Delta\mathbf{U}_1/\Delta\mathbf{I}_1$, is the source impedance for the fundamental frequency, then the active power loss inside of the originally sinusoidal power system is equal to

$$\Delta P_s = P_1 - P + I_1^2\,\mathrm{Re}\,\{\mathbf{Z}_{s1}\}. \tag{4}$$

The quantity x in systems with sinusoidal waveforms of frequency f_1, so that also its crms value, can be determined, according to the Sampling Theorem, by a sequence of uniformly spaced samples x_k of the frequency higher than $2f_1$. The lowest number of samples N in the period T_1, that fulfills the Sampling Theorem at a uniform sampling is $N = 3$. Therefore, from the Discrete Fourier Transform (DFT), used in this paper in the form

$$\mathbf{X}_n = \frac{\sqrt{2}}{N} \sum_{k=0}^{N-1} x_k \, e^{-j2\frac{\pi}{N}nk} \tag{5}$$

we obtain

$$\mathbf{X}_1 = \frac{\sqrt{2}}{3} \sum_{k=0}^{2} x_k \, e^{-j2\frac{\pi}{3}k} \tag{6}$$

so that the complex rms value of a sinusoidal quantity x is determined by three samples (Fig. 1), x_0, x_1, x_2, of $x(t)$ at points $t_k \triangleq 2\frac{\pi}{3}k$, namely

$$\mathrm{Re}\{\mathbf{X}_1\} = \frac{1}{3\sqrt{2}}(2x_0 - x_1 - x_2) \tag{7}$$

$$\mathrm{Im}\{\mathbf{X}_1\} = -\frac{1}{\sqrt{6}}(x_1 - x_2). \tag{8}$$

Thus, the complex rms value of a sinusoidal voltage and current, $\mathbf{U}_1$ and $\mathbf{I}_1$, and hence also the active and reactive powers of a load can be calculated if only three samples u_0, u_1, u_2 of the voltage $u(t)$ at a load terminals and three samples i_0, i_1, i_2 of its current $i(t)$ are known.

Determining the crms value of the fundamental harmonic $\mathbf{X}_1$ of the nonsinusoidal quantity

$$x_0(t) \triangleq X_0 + \sqrt{2}\,\mathrm{Re}\sum_{n=1}^{\infty}\mathbf{X}_n\,\mathrm{e}^{\mathrm{j}n\omega_1 t} \tag{9}$$

involves a number of issues. First of all, the quantity $x_0(t)$ has to be approximated with an M-order trigonometric polynomial

$$x(t) = X_0 + \sqrt{2}\,\mathrm{Re}\sum_{n=1}^{M-1}\mathbf{X}_n\,\mathrm{e}^{\mathrm{j}n\omega_1 t}. \tag{10}$$

The proper choice of order M is of a substantial importance. If a digital method is to be applied for the value $\mathbf{X}_1$ calculation, then M value affects the decision as to the needed number N of samples x_k. This affects the required sampling rate of an A/D converter and the expected aliasing error of the crms $\mathbf{X}_1$ value calculation. It affects also the computation time, dependent on the amount of calculation needed and the hardware used for that. To reduce the hardware requirement at real-time applications, the amount of calculation should be reduced as much as possible.

The Fast Fourier Transform (FFT) algorithms, used generally for the DFT value of the periodic sequences $\{x_k\}$ calculation, provides us with X_0 and all $M-1$ complex rms values $\mathbf{X}_n$ at the same time, so that, they are computationally inefficient, if only the crms value $\mathbf{X}_1$ is needed. Therefore, more efficient methods of this value calculation than FFT algorithms are desirable for some real-time applications.

Such a genuine method is suggested for example by Sawicki [4]. It reduces all calculations to addition of some samples, selected in a strictly specified manner.

Unfortunately, it requires rather a high sampling rate. Namely, the frequency f_c of the sampling clock is equal to the product of all harmonic orders n of harmonics that have to be filtered out, multiplied by two, and by the fundamental frequency, f_1.

For example, if the complex rms value $\mathbf{X}_1$ is to be found in the presence of harmonics of order $n = 2, 3, 5, 7, 11$, then the sampling frequency has to be equal to

$$f_c = 2(2\times3\times5\times7\times11)\,f_1 = 4620 f_1.$$

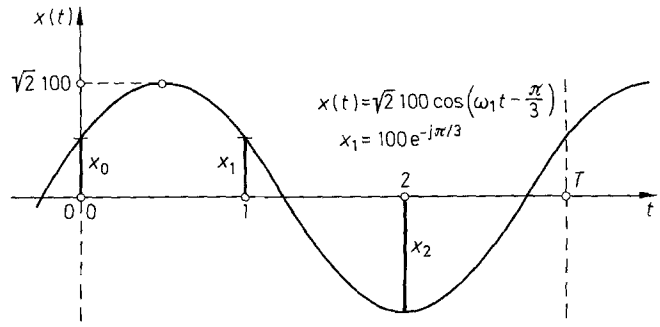

Fig. 1. Example of sinusoidal waveform sampling

This stringent requirement regarding the A/D converter sampling rate may be substantially mitigated if the algorithms discussed in this paper are applied.

2 Algorithm based on the DFT direct calculation

Let the frequency spectrum of the quantity x of frequency f_1 be limited by frequency Mf_1, i.e., $\mathbf{X}_n = 0$ for $n \geqq M$ and let M be an even integer.

It results from the Sampling Theorem that the frequency spectrum of the samples sequence $\{x_k\}$ is not aliased on the condition that the number N of samples taken uniformly in the period T_1 is equal to or higher than $2M$. Let us assume that the samples number N is a multiplicity of four, i.e., $N = 4K$, and that $2K = M$.

In such a case, it results from the DFT that the complex rms value of the fundamental harmonic $\mathbf{X}_1$ is equal to

$$\mathbf{X}_1 = \frac{\sqrt{2}}{N}\sum_{k=0}^{N-1}x_k\,\mathrm{e}^{-\mathrm{j}\frac{2\pi}{N}k} = \frac{\sqrt{2}}{N}\sum_{k=0}^{4K-1}x_k\mathbf{V}^k \tag{11}$$

where

$$\mathbf{V} \triangleq 1\mathrm{e}^{-\mathrm{j}\frac{\pi}{2K}}. \tag{12}$$

Since $\mathbf{V}^{k+M} = -\mathbf{V}^k$ and $\mathbf{V}^0 = 1$, then the DFT formula (11) can be expressed as

$$\mathbf{X}_1 = \frac{\sqrt{2}}{N}\left[(x_0 - x_M) + \sum_{k=1}^{2K-1}(x_k - x_{k+M})\,\mathbf{V}^k\right]. \tag{13}$$

Let us denote $x_k - x_{k+M} \triangleq s_k$. Since $\mathbf{V}^K = -\mathrm{j}1$, formula (13) can be rearranged as follows

$$\mathbf{X}_1 = \frac{\sqrt{2}}{N}\left[s_0 - \mathrm{j}s_K + \sum_{k=0}^{K-1}(s_k\mathbf{V}^k + s_{M-k}\mathbf{V}^{M-k})\right]. \tag{14}$$

The complex coefficients $\mathbf{V}^k$ have a quadrant symmetry, (Fig. 2) i.e.,

$$\mathrm{Re}\{\mathbf{V}^{M-k}\} = -\mathrm{Re}\{\mathbf{V}^k\} \tag{15}$$

$$\mathrm{Im}\{\mathbf{V}^{M-k}\} = \mathrm{Im}\{\mathbf{V}^k\} \tag{16}$$

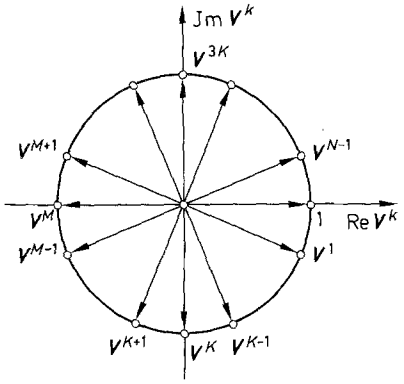

Fig. 2. Quadrant symmetry of $\mathbf{V}^k$ coefficients

hence

$$\mathrm{Re}\,\{\mathbf{X}_1\} = \frac{\sqrt{2}}{N}\left[s_0 + \sum_{k=1}^{K-1}(s_k - s_{M-k})\,\mathrm{Re}\,\{\mathbf{V}^k\}\right] \tag{17}$$

$$\mathrm{Im}\,\{\mathbf{X}_1\} = \frac{\sqrt{2}}{N}\left[-s_k + \sum_{k=1}^{K-1}(s_k + s_{M-k})\,\mathrm{Im}\,\{\mathbf{V}^k\}\right]. \tag{18}$$

This formula for the DFT direct calculation, at the assumption that N is a multiplicity of 4, provides us with a computationally efficient algorithm for the crms value of the fundamental harmonic calculation. If the values of $\mathrm{Re}\,\{\mathbf{V}^k\}$, $\mathrm{Im}\,\{\mathbf{V}^k\}$ and $\sqrt{2}/N$ are stored in the computer memory, then this algorithm requires only $3M - 4$ additions and M multiplications. Example 1. If the quantity x has harmonics of order $n = 0, 1, 2, 3, 4, 5$, then if we assume $N = 12$, $M = 6$, $K = 3$, so that

$$\mathrm{Re}\,\{\mathbf{V}^1\} = \cos(-\pi/6) = \sqrt{3}/2;$$

$$\mathrm{Im}\,\{\mathbf{V}^1\} = \sin(-\pi/6) = -1/2$$

$$\mathrm{Re}\,\{\mathbf{V}^2\} = \cos(-\pi/3) = 1/2;$$

$$\mathrm{Im}\,\{\mathbf{V}^2\} = \sin(-\pi/3) = -\sqrt{3}/2$$

then, we obtain

$$\mathrm{Re}\,\{\mathbf{X}_1\} = \frac{\sqrt{2}}{12}\left[s_0 + \frac{\sqrt{3}}{2}(s_1 - s_5) + \frac{1}{2}(s_2 - s_4)\right] \tag{19}$$

$$\mathrm{Im}\,\{\mathbf{X}_1\} = -\frac{\sqrt{2}}{12}\left[s_3 + \frac{1}{2}(s_1 + s_5) + \frac{\sqrt{3}}{2}(s_2 + s_4)\right]. \tag{20}$$

Calculation of $\mathbf{X}_1$ in such a case requires only 14 additions and 6 multiplications. Moreover, the A/D converter has to provide only $N = 12$ samples in the period T_1 to enable us to calculate the crms value $\mathbf{X}_1$.

For the previously considered case of a quantity with harmonics up to $n = 11$ order, $N = 24$ is the lowest N that fulfills the requirement of the Sampling Theorem and, at the same time, is a multiplicity of 4. Thus, the A/D converter has to provide only 24 samples in the period T_1, so that, a converter with a much lower sampling rate is required now as compared to that needed, if the method suggested in paper [4] is applied.

Apart from the traditional sources of an error which affect samples x_k, and hence $\mathbf{X}_1$ value, the calculation result may be affected by the lack of the sampling synchronization with the frequency of the sampled quantity x and by the spectrum aliasing. These are very peculiar sources of an error, so that, they require a more detailed analysis.

3 The error caused by the lack of sampling synchronization

The samples x_k in formula (11) are synchronous with the sampled quantity $x(t)$. For the sampling frequency, $f_c \triangleq 1/T_c$, equal to an integer multiplicity of the fundamental frequency $f_1 \triangleq 1/T_1$, such that

$$f_c = Nf_1 = 2Mf_1 = 4Kf_1 \tag{21}$$

the samples x_k at points $t_k \triangleq kT_c$, equal to

$$x_k = X_0 + \sqrt{2}\,\mathrm{Re}\sum_{n=1}^{M-1}\mathbf{X}_n\,\mathrm{e}^{\mathrm{j}n\omega_1 t_k} \tag{22}$$

are equidistant and periodic with the discrete period N, so that as k increases from 0 to N the argument $\omega_1 t_k = 2\dfrac{\pi}{N}k$ increases by 2π.

The period T_1 varies randomly in weak power systems, however, even in a range of $\mp 1\%$, and its value at the sampling start is not known. Thus, the sampling frequency f_c is not equal to Nf_1, usually.

The effects of the lack of the sampling frequency synchronization with the sampled quantity can be reduced by its sampling over a sufficiently large number of periods. However, such an approach may increase unacceptably the amount of computation and delay the results. Therefore, it is usually preferable at the real-time applications that the crms $\mathbf{X}_1$ value calculation is based on the data collected in a single period T_1. In such a case the error involved can be evaluated as follows.

If the condition (21) is not fulfilled, then the sequence $\{x_k\}$ is not periodic with the number N, since, as k increases from 0 to N the argument $\omega_1 t_k = 2\dfrac{\pi}{T}T_c k$ does not increase by 2π. Let us suppose that it increases by $2\pi - \vartheta$, so that

$$\omega_1 t_k = \frac{2\pi - \vartheta}{N}k \tag{23}$$

and let us denote the samples at these points by z_k. The samples have the value

$$X_0 + \sqrt{2}\,\mathrm{Re}\sum_{n=1}^{M-1}\mathbf{X}_n\,\mathrm{e}^{\mathrm{j}n\frac{2\pi-\vartheta}{N}k} \triangleq z_k. \tag{24}$$

For small value of the angle ϑ, such that $|n\vartheta| \ll \frac{\pi}{2}$, holds

$$e^{-jn\frac{\vartheta}{N}k} = 1 - jn\frac{\vartheta}{N}k + O^2(jn) \tag{25}$$

where $O^2()$ denotes a small number of the second order. Therefore, the error in the sampling synchronization, described quantitatively with the angle ϑ, changes the true value of the sample x_k to a measured value z_k as follows

$$z_k = x_k - k\frac{\vartheta}{N}\sqrt{2}\,\mathrm{Re}\sum_{n=1}^{M-1} jn\mathbf{X}_n\mathbf{V}^{-nk}. \tag{26}$$

It may be more convenient to describe the lack of synchronization not with the angle ϑ, but with a shift τ of the Nth sample with respect to the period T_1. This shift can be expressed as a part (Fig. 3) of T_c, namely $\tau \triangleq \lambda T_c$. Since for small ϑ, $\vartheta \approx \lambda\frac{2\pi}{N}$, we have

$$z_k = x_k - \lambda k\frac{2\pi}{N^2}\sqrt{2}\,\mathrm{Re}\sum_{n=1}^{M-1} jn\mathbf{X}_n\mathbf{V}^{-nk}. \tag{27}$$

The samples z_k, used in the DFT procedure for the crms $\mathbf{X}_1$ value calculation instead of x_k, do not result in this value, but in

$$\frac{\sqrt{2}}{N}\sum_{k=0}^{N-1} z_k\mathbf{V}^k \triangleq \mathbf{Z}_1. \tag{28}$$

Formula (27) shows that the synchronization error λ changes x_k by an additive component, proportional to λ and to the crms value $\mathbf{X}_n$ of each individual harmonic separately. Therefore, we can analyze separately the effect of a single harmonic on the crms $\mathbf{X}_1$ value calculation. Let us denote

$$\mathbf{X}_n \triangleq X_n\mathbf{1}_n, \quad \text{where} \quad \mathbf{1}_n \triangleq 1e^{j\xi_n} \tag{29}$$

and assume that the quantity x contains only one harmonic of the crms value $\mathbf{X}_n$. In such a case

$$z_k = x_k + X_n\lambda n\frac{2\pi}{N^2}k\sqrt{2}\,\mathrm{Im}\{\mathbf{1}_n\mathbf{V}^{-nk}\} \tag{30}$$

so that the sequence $\{z_k\}$ is transformed with the DFT procedure into

$$\mathbf{Z}_1 = \frac{\sqrt{2}}{N}\sum_{k=0}^{N-1}\left[x_k + \lambda X_k n\frac{2\pi}{N^2}k\sqrt{2}\,\mathrm{Im}\{\mathbf{1}_n\mathbf{V}^{-nk}\}\right]\mathbf{V}^k$$
$$= \mathbf{X}_1 + \lambda\mathbf{S}_n X_n \tag{31}$$

where

$$\mathbf{S}_n \triangleq 4\pi\frac{n}{N^3}\sum_{k=1}^{N-1} k\,\mathrm{Im}\{\mathbf{1}_n\mathbf{V}^{-nk}\}\,\mathbf{V}^k. \tag{32}$$

On the condition that the value of X_n can be anticipated, the equation (31) enables us to evaluate the error in the crms value $\mathbf{X}_1$ calculation caused by the lack of the sampling frequency f_c synchronization with the frequency f_1 of the sampled quantity.

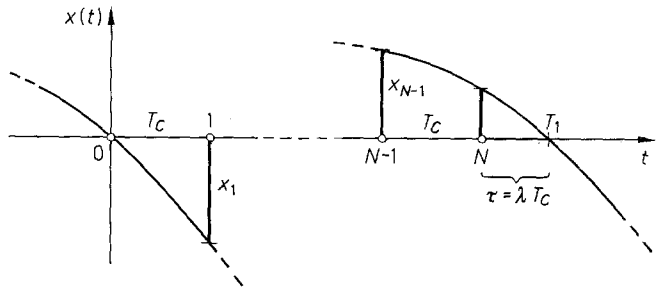

Fig. 3. Asynchronous sampling

The impact of particular harmonics on the $\mathbf{X}_1$ calculation error is specified by the complex coefficient $\mathbf{S}_n$. At a chosen number of samples N, it depends on the harmonic order n and its phase angle ξ_n. This angle is not known usually, however. Therefore, only the maximum value of the real and imaginary parts of $\mathbf{S}_n$ can be calculated. These parts are equal to

$$\mathrm{Re}\{\mathbf{S}_n\} = 4\pi\frac{n}{N^3}\sum_{k=1}^{N-1} k\sin\left(\xi_n + 2\frac{\pi}{N}nk\right)\cos\left(2\frac{\pi}{N}k\right) \tag{33}$$

$$\mathrm{Im}\{\mathbf{S}_n\} = -4\pi\frac{n}{N^3}\sum_{k=0}^{N-1} k\sin\left(\xi_n + 2\frac{\pi}{N}nk\right)\sin\left(2\frac{\pi}{N}k\right). \tag{34}$$

Their maximum value for some N and n are tabulated in Table 1.

The formula (31) and the values tabulated in Table 1 enable us to evaluate a worse-case impact of particular harmonics on the result of the crms value $\mathbf{X}_1$ calculation at the lack of the sampling frequency synchronization, specified with the coefficient λ.

Table 1. Some values of the $\mathbf{S}_n$ coefficient

n	\	$N = 8$	$N = 12$	$N = 28$	$N = 64$
1.	Re $\{\mathbf{S}_1\}$	0.30	0.22	0.105	0.048
	Im $\{\mathbf{S}_1\}$	0.39	0.26	0.112	0.050
2.	Re $\{\mathbf{S}_2\}$	0.34	0.22	0.094	0.041
	Im $\{\mathbf{S}_2\}$	0.20	0.12	0.048	0.021
3.	Re $\{\mathbf{S}_3\}$	0.33	0.20	0.081	0.035
	Im $\{\mathbf{S}_3\}$	0.15	0.075	0.028	0.012
4.	Re $\{\mathbf{S}_4\}$		0.21	0.078	0.034
	Im $\{\mathbf{S}_4\}$		0.064	0.020	0.008
5.	Re $\{\mathbf{S}_5\}$		0.23	0.077	0.033
	Im $\{\mathbf{S}_5\}$		0.063	0.016	0.007
6.	Re $\{\mathbf{S}_6\}$			0.078	0.033
	Im $\{\mathbf{S}_6\}$			0.014	0.005
7.	Re $\{\mathbf{S}_7\}$			0.080	0.033
	Im $\{\mathbf{S}_7\}$			0.013	0.005
14.	Re $\{\mathbf{S}_{14}\}$				0.034
	Im $\{\mathbf{S}_{14}\}$				0.003
15.	Re $\{\mathbf{S}_{15}\}$				0.034
	Im $\{\mathbf{S}_{15}\}$				0.003

4 Aliasing error

Another source of an error at the crms $\mathbf{X}_1$ value calculation, peculiar for digital methods, is the spectrum aliasing phenomenon. It occurs, if the doubled frequency $2f_n$ of some harmonics of the sampled quantity x_0, is higher than the sampling frequency f_c.

Its effect on the crms value $\mathbf{X}_1$ of the fundamental harmonic calculation can be expressed as follows.

Let the spectrum $\{\mathbf{X}_n\}$ of x be not limited as to the frequency band, then

$$x_0 = X_0 + \sqrt{2}\,\mathrm{Re}\sum_{n=1}^{\infty}\mathbf{X}_n\,\mathrm{e}^{\mathrm{j}n\omega_1 t} = \frac{1}{\sqrt{2}}\sum_{-\infty}^{\infty}\mathbf{X}_n\,\mathrm{e}^{\mathrm{j}n\omega_1 t} \tag{35}$$

with

$$\mathbf{X}_{-n} = \mathbf{X}_n^*. \tag{36}$$

If x_0 is sampled with the frequency $f_c = Nf_1$, then

$$x_k = \frac{1}{\sqrt{2}}\sum_{n=-\infty}^{\infty}\mathbf{X}_n\mathbf{V}^{-nk}. \tag{37}$$

The N-point DFT of the sequence $\{x_k\}$, used for the crms value $\mathbf{X}_1$ calculation results in

$$\mathbf{Z}_1 = \frac{\sqrt{2}}{N}\sum_{k=0}^{N-1}x_k\mathbf{V}^k = \frac{\sqrt{2}}{N}\sum_{k=0}^{N-1}\left[\frac{1}{\sqrt{2}}\sum_{n=-\infty}^{\infty}\mathbf{X}_n\mathbf{V}^{-nk}\right]\mathbf{V}^k$$
$$= \frac{1}{N}\sum_{n=-\infty}^{\infty}\mathbf{X}_n\sum_{k=1}^{N-1}\mathbf{V}^{k(1-n)}. \tag{38}$$

Since

$$\sum_{k=0}^{N-1}\mathbf{V}^{k(1-n)} = \sum_{k=0}^{N-1}\mathrm{e}^{-\mathrm{j}2\frac{\pi}{N}k(1-n)} = \begin{cases}0, & \text{for } 1-n \neq rN\\ N, & \text{for } 1-n = rN\end{cases} \tag{39}$$

with $r = 0, 1, 2, 3, \ldots$, thus, the right side of formula (38) has a nonzero value only for $n = 1 \pm rN$. Hence

$$\mathbf{Z}_1 = \mathbf{X}_1 + \sum_{r=1}^{\infty}(\mathbf{X}_{1+rN} + \mathbf{X}_{1-rN}) \tag{40}$$

or, taking into account the relation (36)

$$\mathbf{Z}_1 = \mathbf{X}_1 + \sum_{r=1}^{\infty}(\mathbf{X}_{rN-1}^* + \mathbf{X}_{rN+1}) \tag{41}$$

so that, each harmonic of order $rN \pm 1$, $r = 1, 2, 3, \ldots$ disturbs the result of $\mathbf{X}_1$ calculation with the N-point DFT by the crms value $\mathbf{X}_{rN-1}^*$ or $\mathbf{X}_{rN+1}$. Thus, the aliasing error, $\mathbf{A}_N$, of the crms $\mathbf{X}_1$ value calculation with N-point DFT is equal to

$$\mathbf{A}_N \triangleq \mathbf{Z}_1 - \mathbf{X}_1 = \sum_{r=1}^{\infty}(\mathbf{X}_{rN-1}^* + \mathbf{X}_{rN+1}). \tag{42}$$

Observe, that just characteristic harmonics of the power systems may contribute to this error. For example, if $N = 12$, then

$$\mathbf{A}_{12} = \mathbf{X}_{11}^* + \mathbf{X}_{13} + \mathbf{X}_{23}^* + \mathbf{X}_{25} + \ldots \tag{43}$$

so that, the aliasing error is composed of harmonics crms $\mathbf{X}_n$ that are expected to have, usually, a higher value than the remaining ones.

5 Algorithm based on the cyclic convolution

An algorithm of the crms value $\mathbf{X}_1$ of the fundamental harmonic calculation can be based also on the cyclic convolution. Before it is developed, let us observe the following interesting relation between some samples of a sinusoidal quantity and its crms value.

Let us express a sinusoidal waveform $z(t)$ of f_1 frequency as follows

$$z(t) = \sqrt{2}\,Z_1\cos(\omega_1 t + \alpha_1) = \sqrt{2}\,Z_1\cos\alpha_1\cos\omega_1 t$$
$$-\sqrt{2}\,Z_1\sin\alpha_1\cos\left(\omega_1 t - \frac{\pi}{2}\right). \tag{44}$$

If $z(t)$ is sampled with the frequency $f_c = Nf_1 = 4Kf_1$, then

$$z_k = \sqrt{2}\,Z_1\cos\alpha_1\cos\frac{\pi k}{2K}$$
$$-\sqrt{2}\,Z_1\sin\alpha_1\cos\left(\frac{\pi k}{2K} - \frac{\pi}{2}\right) \tag{45}$$

so that

$$z_0 = \sqrt{2}\,Z_1\cos\alpha_1,$$
$$z_K = -\sqrt{2}\,Z_1\sin\alpha_1. \tag{46}$$

Since $\mathbf{Z}_1 \triangleq Z_1\,\mathrm{e}^{\mathrm{j}\alpha_1} = Z_1\cos\alpha_1 + \mathrm{j}Z_1\sin\alpha_1$, hence the crms value $\mathbf{Z}_1$ can be expressed as

$$\mathbf{Z}_1 = \frac{1}{\sqrt{2}}(z_0 - \mathrm{j}z_K) \tag{47}$$

thus, to determine the crms value $\mathbf{Z}_1$ of a sinusoidal quantity, it is enough to know only two of its samples, z_0 and z_K, taken at $t = 0$ and $t = T_1/4$.

Now, let us assume that the periodic quantity x does not contain any harmonic of order n equal to or higher than $N - 1$. Let its samples x_k, with the period N, form the input sequence $\{x_k\}$ of a digital filter which attenuates entirely all harmonics, apart from the fundamental one which remains unchanged. It means, that the Discrete Fourier Transform $\mathbf{H}_n$ of the filter impulse response h_k has to have the values

$$\mathbf{H}_n \triangleq H_n\,\mathrm{e}^{\mathrm{j}\eta_n} \triangleq \begin{cases}1, & \text{for } n = 1,\ N-1\\ 0, & \text{for } n = 0, 2, 3, \ldots, N-2.\end{cases} \tag{48}$$

Such a filter passes only the samples of the fundamental harmonic of the input sequence $\{x_k\}$, i.e., the samples z_k of sinusoidal waveform. The output samples z_k of the filter can be calculated with the cyclic convolution [6] of the filter impulse response h_k and the samples x_k, namely

$$z_k = \sum_{n=0}^{N-1}h_{\langle k-n\rangle}x_n \tag{49}$$

where

$$h_k \triangleq \frac{1}{N} \sum_{n=0}^{N-1} \mathbf{H}_n \, e^{j2\frac{\pi}{N}nk} = \frac{2}{N} \cos\left(2 \frac{\pi}{N} k\right) \tag{50}$$

and the index $\langle s \rangle$ has the following meaning

$$\langle s \rangle \triangleq \begin{cases} s, & \text{for } s \geqq 0 \\ N - s, & \text{for } s < 0. \end{cases} \tag{51}$$

Thus, with $h_k' \triangleq h_k/\sqrt{2}$,

$$\mathbf{X}_1 = \mathbf{Z}_1 = \sum_{n=0}^{N-1} (h'_{\langle -n \rangle} - j h'_{\langle K-n \rangle}) \, x_n. \tag{52}$$

This algorithm requires more computation than that discussed in Sec. 2, but still less than the FFT procedures. If the values h_k' are stored in a look-up table, then the number of multiplications required for $\mathbf{X}_1$ calculation is equal to $2N$. By processing N samples of x, this algorithm eliminates the harmonics of order n up to $N - 2$, however, while the previous algorithm only up to $N/2 - 1$.

It is property of the cyclic convolution (52) that if $x(t)$ contains harmonics of order $n = rN \pm 1$, $r = 1, 2, \ldots$, then, at the filter output their samples can not be distinguished from the samples of the fundamental harmonic. Thus, they affect the values of z_0 and z_K samples, so that also the crms value $\mathbf{X}_1$ computation result.

Fortunately, this effect of the spectrum aliasing can be mitigated now easier than it was possible with the algorithm discussed previously. It imposes weaker requirement as regard to the samples number N. Previously, it had to be a multiplicity of 4.

Now, any number of samples can be processed with the cyclic convolution. If N is chosen to be between the orders n of two neighbouring harmonics that are expected to have a low rms value, then the aliasing error may be limited without any increase in the sampling rate. Harmonics of even order are, usually, much lesser than those of order $6n \pm 1$ in three-phase systems.

If, for instance, N is chosen to be 13, i.e., between the 12th and 14th harmonics, then the algorithm discussed is able to provide us with the crms value $\mathbf{X}_1$ of a quantity that contains up to 24th harmonic, with the aliasing error composed of only the crms values $\mathbf{X}_{12}$ and $\mathbf{X}_{14}$, the small ones, usually. This requires only 13 samples per period and 26 multiplications, i.e., approximately 1/2 sample and 2 multiplications per harmonic contained in the sampled quantity.

If N is not divisible by 4, however, than the output sample z_K does not exist. Nonetheless, this sample can be calculated after the sample x_M, for $M \triangleq N$ div 4, is obtained with the cyclic convolution. Namely,

$$z_M = \sum_{n=0}^{N-1} h_{\langle M-n \rangle} x_n. \tag{53}$$

If this sample and the sample z_0 are known, then it results from the formula (45), that

$$\begin{aligned} z_M &= \sqrt{2}\, Z_1 \cos\alpha_1 \cos\left(2 \frac{\pi}{N} M\right) \\ &\quad - \sqrt{2}\, Z_1 \sin\alpha_1 \cos\left(2 \frac{\pi}{N} M - \frac{\pi}{2}\right) \\ &= z_0 \cos\left(2 \frac{\pi}{N} M\right) + z_K \cos\left(2 \frac{\pi}{N} M - \frac{\pi}{2}\right) \end{aligned} \tag{54}$$

hence

$$z_K = \frac{z_M - z_0 \cos\left(2 \frac{\pi}{N} M\right)}{\sin\left(2 \frac{\pi}{N} M\right)}. \tag{55}$$

Thus, the sample z_K can be found even if N is not divisible by 4.

6 Conclusions

The algorithms of the complex rms value $\mathbf{X}_1$ of the fundamental harmonic calculation discussed in this paper are very effective computationally and they require low sampling rate of the A/D converter. They are much more suitable for the real-time applications than those based on the FFT algorithms. Of the two algorithms considered, that derived from the DFT, requires an exceptionally low number of multiplications, while that based on the cyclic convolution, enables for the aliasing error reduction by a proper choice of the samples number N, without increasing the sampling rate of the A/D converter.

References

1. E. W. Kimbark: Direct Current Transmission, Vol. 1, Wiley-Interscience, 1971
2. W. Depenbrock: Wirk- und Blindleistungen periodischer Ströme in Ein- und Mehrphasensystemen mit periodischen Spannungen beliebiger Kurvenform, ETG-Fachtagung „Blindleistung", Aachen, October 1979.
3. L. S. Czarnecki: Measurement of Internal Loss of Active Power in Sinusoidal Power Systems with Non-Linear and Time-Variant Loads, Proc. of the IMEKO TC4 Symposium "Measurements in Electrical and Electronic Power Systems", Zurich, 1989
4. J. Sawicki: Theoretische Grundlagen der speziellen Samplings-Methode zur Messung der komplexen Form einer Grundschwingung, Archiv für Elektrotechnik, 71 (1988), pp. 337–348
5. A. V. Oppenheim and R. W. Schafer: Digital Signal Processing, Prentice Hill, Inc, 1975
6. C. S. Burrus and T. W. Parks: DFT/FFT and Convolution Algorithms, Theory and Implementation, John Wiley & Sons, 1985

Received: July 17, 1991

Dr. habil. Leszek S. Czarnecki
Electrical and Computer Engineering
Louisiana State University
Baton Rouge, LA 70803, USA